

MTH 346/646 - Sixteenth day of class - 10/21

Fun fact: Take an elliptic curve E and a point $P \in E(\mathbb{Q})$ of infinite order. For each prime p (for which $E/\mathbb{F}_p$ is non-singular) $\phi_p(P)$ has finite order.

Q: How often is the order odd? More precisely, what is

$$\lim_{x \rightarrow \infty} \frac{\text{number of primes } p \leq x \text{ so that } \phi_p(P) \text{ has odd order}}{\text{number of primes } p \leq x}?$$

A: Most of the time, this limit equals $\frac{11}{21}$.

(This happens, for example, with $E : y^2 = x^3 - 16x + 16$ and $P = (0, 4)$.)

Last time: Mordell's theorem and how the finite generation of $C(\mathbb{Q})$ follows from four lemmas. We proved the second lemma.

Today: Proof of Lemma 3.

Proof of Lemma 3

Recall that $C : y^2 = x^3 + ax^2 + bx + c$ with $a, b, c \in \mathbb{Z}$. If $P = (x, y)$ is a rational point on $C(\mathbb{Q})$, then $x = \frac{m}{e^2}$ where $\gcd(m, e) = 1$ and we define $h(P) = \log \max\{|m|, e^2\}$.

Lemma 3. *There is a constant κ , depending on a, b and c so that*

$$h(2P) \geq 4h(P) - \kappa \quad \forall P \in C(\mathbb{Q}).$$

Again, we can prove this only for the points P outside of some finite set S . We let $P = (x, y)$ and $2P = (\xi, \eta)$. The duplication formula gives that

$$\xi = \frac{x^4 - 2bx^2 - 8cx + b^2 - 4ac}{4x^3 + 4ax^2 + 4bx + 4c}.$$

Setting $x = m/e^2$ gives

$$\xi = \frac{m^4 - 2bm^2e^2 - 8cme^6 + (b^2 - 4ac)e^8}{4m^3 + 4am^2e^2 + 4bme^4 + 4ce^6}.$$

Goal: Show that there is a bounded amount of cancellation between the numerator and the denominator.

Lemma (Lemma 3'). *Let $\phi(X)$ and $\psi(X)$ be polynomials with integer coefficients and no common (complex) roots. Let d be the maximum of the degrees of ϕ and ψ .*

(a) *There is an integer $R \geq 1$, depending on ϕ and ψ , so that for all rational numbers $\frac{m}{n}$,*

$$\gcd\left(n^d \phi\left(\frac{m}{n}\right), n^d \psi\left(\frac{m}{n}\right)\right)$$

divides R .

(b) There are constants κ_1 and κ_2 , depending on ϕ and ψ , so that for all rational numbers $\frac{m}{n}$ which are not roots of ψ ,

$$dh\left(\frac{m}{n}\right) - \kappa_1 \leq h\left(\frac{\phi(m/n)}{\psi(m/n)}\right) \leq dh\left(\frac{m}{n}\right) + \kappa_2.$$

This will prove Lemma 3. We will use part (b) with $\phi = x^4 - 2bx^2 - 8cx + b^2 - 4ac$ and $\psi = 4x^3 + 4ax^2 + 4bx + 4c$.

Problem 3(a) on homework 6 implies that there are polynomials with integer coefficients $F(x)$ and $\Phi(x)$ with integer coefficients so that

$$F(x)\psi(x) + \Phi(x)\phi(x) = 4D.$$

This implies that ϕ and ψ have no common complex roots.

Then we use part (b) with $n = e^2$ so $m/n = m/e^2 = x$. We get

$$4h(P) - \kappa_1 \leq h(2P) \leq 4h(P) + \kappa_2.$$

Proof of Lemma 3': Assume that $\deg(\phi) = d$ and $\deg(\psi) = e \leq d$.

(a)

Then since ϕ and ψ both have degree at most d , the quantities $n^d\phi\left(\frac{m}{n}\right)$ and $n^d\psi\left(\frac{m}{n}\right)$ are both integers.

Since $\phi(X)$ and $\psi(X)$ have no common complex root, when we compute the greatest common divisor of them (using the Euclidean algorithm via polynomial long division), we find that $\gcd(\phi(X), \psi(X)) = 1$. Doing this process backwards gives us a linear combination of $\phi(X)$ and $\psi(X)$ that equals 1. We get

$$F(X)\phi(X) + G(X)\psi(X) = 1,$$

where $F(X)$ and $G(X)$ are polynomials with rational coefficients. Multiply through by A , the LCM of the denominators of coefficients of F and G to get

$$AF(X)\phi(X) + AG(X)\psi(X) = A.$$

Let D be the maximum of the degrees of F and G . We then get

$$An^{d+D}F\left(\frac{m}{n}\right)\phi\left(\frac{m}{n}\right) + An^{d+D}G\left(\frac{m}{n}\right)\psi\left(\frac{m}{n}\right) = An^{d+D}.$$

Let $\Phi(m, n) = n^d\phi\left(\frac{m}{n}\right)$ and $\Psi(m, n) = n^d\psi\left(\frac{m}{n}\right)$ and $\gamma = \gcd(\Phi(m, n), \Psi(m, n))$.

Claim: Let a_0 be the constant coefficient of $\phi(X)$. Then $\gamma | Aa_0^{D+d}$.

Proof: If $\psi(X) = a_0X^d + a_1X^{d-1} + \dots$, then

$$An^{D+d-1}\Phi(m, n) = Aa_0m^dn^{D+d-1} + Aa_1m^{d-1}n^{D+d} + \dots$$

The first term is $Aa_0m^dn^{D+d-1}$, while all other terms are multiples of An^{D+d} . Since γ divides $\Phi(m, n)$, and γ divides An^{D+d} , it follows that γ divides $Aa_0m^dn^{D+d-1}$.

Thus, γ divides $\gcd(An^{D+d}, Aa_0m^dn^{D+d-1})$.

Now, recall that $\gcd(m, n) = 1$. Now let $\delta = \gcd(\gamma, A)$ and write $\gamma = \delta\epsilon$. It follows that ϵ divides $\gcd(n^{D+d}, a_0m^dn^{D+d-1})$. Since $\gcd(m, n) = 1$, because $\epsilon|n^{D+d}$ we have that $\gcd(\epsilon, m) = 1$ and so ϵ divides a_0n^{D+d-1} . It follows that $\gamma = \delta\epsilon$ divides Aa_0n^{D+d-1} .

So, at the expense of multiplying by a_0 , we can decrease the power on n by 1. We can play the same game with

$$Aa_0n^{D+d-2}\Phi(m, n)$$

and conclude that γ divides $Aa_0^2n^{D+d-2}$. We keep going, and eventually get that γ divides Aa_0^{D+d} .

This proves part (a) with $R = Aa_0^{D+d}$.

(b) We will only prove the lower bound, because this is what is needed for the proof of Lemma 3. The upper bound can be proven using techniques similar to those in the proof of Lemma 2.

As usual, we will exclude a finite number of rational numbers m/n . These would merely adjust the value of κ_1 .

Since $h(r) = h(1/r)$, we can assume (by reversing roles of ϕ and ψ) that $\deg(\phi) = d$ and $\deg(\psi) = e \leq d$.

Let $m, n \in \mathbb{Z}$ with $\gcd(m, n) = 1$ and let

$$\xi = \frac{\phi(m/n)}{\psi(m/n)} = \frac{n^d\phi(m/n)}{n^d\psi(m/n)} = \frac{\Phi(m, n)}{\Psi(m, n)}.$$

If we let $d = \gcd(\Phi(m, n), \Psi(m, n))$, then if we divide out by this factor of d we are left with a reduced fraction. So

$$H(\xi) = \max\{|\Phi(m, n)|/d, |\Psi(m, n)|/d\}.$$

By part (a), we know that there is a positive integer R so that $\gcd(\Phi(m, n), \Psi(m, n))|R$. Now, we have

$$\begin{aligned} H(\xi) &\geq \frac{1}{R} \max\{|\Phi(m, n)|, |\Psi(m, n)|\} \\ &\geq \frac{1}{R} \max\{|n^d\phi(m/n)|, |n^d\psi(m/n)|\} \\ &\geq \frac{1}{2R} (|n^d\phi(m/n)| + |n^d\psi(m/n)|). \end{aligned}$$

Now, we look at the function $\frac{H(\xi)}{H(m/n)^d}$. We have

$$\begin{aligned}\frac{H(\xi)}{H(m/n)^d} &\geq \frac{1}{2R} \cdot \frac{|n^d \phi(m/n)| + |n^d \psi(m/n)|}{\max\{|m|^d, |n|^d\}} \\ &= \frac{1}{2R} \cdot \frac{|\phi(m/n) + \psi(m/n)|}{\max\{|m/n|^d, 1\}}.\end{aligned}$$

Define

$$p(t) = \frac{|\phi(t)| + |\psi(t)|}{\max\{|t|^d, 1\}}.$$

This function is continuous. Since one of $\deg \phi$ or $\deg \psi = d$, we have that $\lim_{t \rightarrow \infty} p(t)$ exists and is positive, and the same with $\lim_{t \rightarrow -\infty} p(t)$.

Also $p(t)$ is never zero, because if $p(t) = 0$, then t is a common real root of ϕ and ψ . Thus, the absolute minimum of $p(t)$ is positive. Thus there is some $C_1 > 0$ so $p(t) \geq C_1$ for all real numbers t .

Unpacking the inequalities gives $H(\xi) \geq \frac{C_1}{2R} H(m/n)^d$, and taking logs gives

$$h(\xi) \geq dh(m/n) - \kappa_1$$

where $\kappa_1 = \log(2R/C_1)$.

QED Lemma 3' and Lemma 3

Overview of Lemma 4

Step 1:

We will only prove Lemma 4 for an elliptic curve with a rational point of order 2. If we move the point of order 2 to $T = (0, 0)$, the curve takes the form $C : y^2 = x^3 + ax^2 + bx$. This curve has a “friend”

$$\overline{C} : y^2 = x^3 + \overline{a}x^2 + \overline{b}x,$$

where $\overline{a} = -2a$ and $\overline{b} = a^2 - 4b$.

There are morphisms $\phi : C \rightarrow \overline{C}$ and $\overline{\phi} : \overline{C} \rightarrow C$ so that for any $P \in C(\mathbb{Q})$,

$$\overline{\phi}(\phi(P)) = 2P$$

and for any $\overline{P} \in \overline{C}(\mathbb{Q})$,

$$\phi(\overline{\phi}(\overline{P})) = 2\overline{P}.$$

This means that a point $P \in C(\mathbb{Q})$ is twice a point (that is, in $2C(\mathbb{Q})$) if and only if it can be written in the form $P = \overline{\phi}(Q)$ for some $Q \in \overline{C}(\mathbb{Q})$ and Q can be written in the form $Q = \phi(R)$ for some $R \in C(\mathbb{Q})$.

Q1: What are the kernels of ϕ and $\overline{\phi}$?

A: $\ker \phi = \{\mathcal{O}, T\}$ and $\ker \overline{\phi} = \{\overline{\mathcal{O}}, \overline{T}\}$.

Q2: What are the images of ϕ and $\bar{\phi}$?

A: The image of $\bar{\phi}$ is the set of points in $C(\mathbb{Q})$ for which the x -coordinate is the square of a nonzero rational number (most of time T is not in the image of ϕ , but it is when b is a square). Same thing with ϕ .

Step 2: Two more homomorphisms.

Let $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$ be the set of nonzero rational numbers under multiplication mod squares. There is a homomorphism $\alpha : C(\mathbb{Q}) \rightarrow (\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$ defined by taking $\alpha((x, y)) = x \cdot (\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$.

Similarly, there is a homomorphism $\bar{\alpha} : \bar{C}(\mathbb{Q}) \rightarrow (\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$.

From the answers above, we have that $\ker \alpha = \text{im } \bar{\phi}$ and $\ker \bar{\alpha} = \text{im } \phi$.

Theorem: The image of α is a finite subgroup of $(\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$, similarly, the image of $\bar{\alpha}$ is a finite subgroup of $(\mathbb{Q}^\times)/(\mathbb{Q}^\times)^2$.

Calculation: $[C(\mathbb{Q}) : 2C(\mathbb{Q})] \leq |\text{im } \alpha| \cdot |\text{im } \bar{\alpha}|$.

Note: The proof of the theorem above will show that we can put an explicit bound on $|\text{im } \alpha|$ that depends on the elliptic curve C . However, mathematicians do not know if there is an algorithm to compute $\text{im } \alpha$ exactly.